

Blessed Madukoma

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Software and Research Engineer

Summary

Software engineer and systems researcher building reliable infrastructure for AI-generated workloads, with Carnegie Mellon research on whether source data meets the requirements of agent-generated programs.

Professional Experience

- Graduate Research Associate** June 2026 – Present
Carnegie Mellon University, Spatial and Languages Technologies (SaLT) Lab Kigali, Rwanda
 - Studying how to make AI-generated work over tabular data more reliable: whether a low-latency gate can derive a program's exact data requirements and check its source before execution.
 - Examined 12,246 agent-written code cells from 15 public DABstep submissions and tested a check for columns missing from an agent's chosen CSV. In six submissions with recorded errors, it flagged 53 error cells missed by an undefined-name check, but also 156 cells without a recorded error.
- Graduate Research Assistant** Sep 2025 – May 2026
Carnegie Mellon University, Spatial and Languages Technologies (SaLT) Lab Kigali, Rwanda
 - Built an athletics anomaly-detection platform (FastAPI, PostgreSQL & DuckDB, Redis) over 1.5M+ performances from 19,000+ competitions, with an investigator dashboard and audit trails to the underlying performance and sanctions data. [Live platform](#).
 - Implemented 14 detection methods (statistical, ML, and Bayesian) with explainable consensus flagging, elite-filtered thresholds, and end-to-end audit trails; published at IEEE SDS 2026 (arXiv:2604.21953), receiving [Best Paper Honorable Mention](#).
- Data Engineering & Analytics Engineer (Contract)** May 2025 – Sep 2025
Ministry of Infrastructure (MININFRA) Kigali, Rwanda
 - Architected Rwanda's first [infrastructure intelligence platform](#), consolidating siloed construction permit and land-use datasets into a centralized system enabling evidence-based urban planning.
 - Built end-to-end data pipelines for construction permits and land-use records, establishing 12+ urban development indicators including permit throughput, and capital allocation trends.
- Software Engineer (MSc Internship)** Jun 2025 – Sep 2025
Huza HR Kigali, Rwanda
 - Built email notification system with multi-stage recruitment templates, automating candidate communications, recruiter alerts, and internal notifications across the hiring pipeline, enabling the company's first fully digital ATS workflow.
 - Implemented core recruitment workflows using Next.js and Node.js, conducting 5+ peer code reviews to establish scalable architecture that onboarded 50+ job seekers and 10+ recruiters within first deployment.
- Software Engineer** Sep 2024 – Dec 2024
The UPANZI Network Kigali, Rwanda
 - Migrated African [datasets hosting platform](#) to Tailwind CSS, reducing bundle size by 30%, improving customization by 25%, and ensuring data privacy standards while maintaining sensitive datasets.
 - Collaborated with cross-functional teams using Agile methods and Git-based version control system, fostering a 15% increase in team productivity and ensuring timely project delivery.
- Golang Backend Engineer** Mar 2024 – Aug 2024
Platnova Remote
 - Implemented CRON-driven vault settlement automation, reducing payment processing time 20% and increasing transaction speed 40%.
 - Revamped transaction receipt system, increasing customer satisfaction 30% and reducing support inquiries 15%.
 - Executed lifestyle-stays package launch, driving customer engagement and increasing revenue by 15%.
- Software Architect (National Youth Service)** Dec 2022 – Nov 2023
21st Century Technologies Lagos, Nigeria

- Co-led the architecture and delivery of large-scale applications, from system design through implementation with the engineering team.
- Built CI/CD pipelines that reduced release time by 40%; supported DNS restoration and failover for services in Tier III/IV data centres.
- Built data pipelines and reporting dashboards that gave management access to operational information.

Education

• Carnegie Mellon University

MSc. Information Technology — Software Engineering, Big Data & Applied ML

- **Thesis Focus:** Performance anomaly detection in sports using ML models and statistical methods with systems engineering [application](#). *Advisor: Prof. Prasenjit Mitra.*

• Babcock University

BSc. Software Engineering — First Class Honors

Technical Projects

• Sedge [\[Live\]](#)

Tools: Go, PostgreSQL, Redis

- Designed and deployed Sedge's Go backend and data pipelines using PostgreSQL and Redis for a platform covering pipelines, governance, and analysis.
- Prototyping checks for AI-generated SQL with GoSQLX: extracting referenced tables and columns and comparing them with source schemas before execution.

• Performance Anomaly Detection in Athletics [\[Live\]](#)

Carnegie Mellon University Africa / 2025–2026

Tools: Python (FastAPI), PostgreSQL, DuckDB, Redis, HTMX

- Built a platform for analysing 1.5M+ athletics performances from 19,000+ competitions (2010–2025), with an investigator dashboard for reviewing flagged performances and their underlying records.
- Compared 14 statistical, ML, and Bayesian detection methods; added anomaly drill-down, athlete profiles, audit trails, and comparisons with Athletics Integrity Unit sanctions records.

Technical Skills & Certifications

- **Languages & Core:** Go, Python, TypeScript, JavaScript, SQL
- **Backend & Data Systems:** FastAPI, Node/Express, PostgreSQL, Redis, ETL, Data Modeling
- **ML & Analytics:** Scikit-learn, Time Series Analysis, Statistical Modeling
- **Cloud & DevOps:** AWS, Docker, Git/GitHub, Bitbucket, CI/CD (GitHub Actions)
- **Architecture:** Distributed Systems, Microservices, Event-Driven Architecture
- **Frontend:** Vue.js, Nuxt
- **Certifications:** [Data Engineering Essentials](#), [Data Engineering Introduction](#), [Docker](#), [Agile with Jira](#)

Publications

- [1] Madukoma, B., Mitra, P. (2026). **Performance Anomaly Detection in Athletics: A Benchmarking System with Visual Analytics.** *IEEE Swiss Conference on Data Science and AI (SDS2026), Poster Presentation.* Zurich, Switzerland. **Best Paper Honorable Mention.** [ArXiv](#).
- [2] Wafula, S., Madukoma, B. (2025). **Unmasking COVID-19 Vulnerability in Nigeria: Mapping Risks Beyond Urban Hotspots.** *Women in Machine Learning (WiML) Workshop @ NeurIPS 2025.* [OpenReview](#).

Relevant Coursework

- **Machine Learning in Production (Prof. Christian Kästner):** Production AI systems, evaluation, MLOps, and responsible AI.
- **Engineering Data-Intensive Scalable Systems (Paulo Merson):** Distributed system design, streaming systems, containers, and observability.

Leadership & Achievements

- **Chairman** — Academic Innovation Sub-Committee, Research Committee, CMU Africa (2024–2026).
- **Executive President** — Computer Club, Babcock University (2022). Led 2k+ tech member community.
- **Volunteer** — Google DevFest Kigali (Dec 2024).
- **Participant** — Digital Humanism Summer School, CMU-Africa (2025).